(RELEASE 1)

C++ EQUALIZATOR PLUG-IN: USER'S MANUAL (ENG)

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1. FACILITY

This first version (Release 1) is distributed in two formats:

- **VST3 Version (Plugin):** For use within DAWs (Ableton, Reaper, FL Studio, etc.).
  Copy the .vst3 file to C:\Program Files\Common Files\VST3
- **Standalone version (Executable .exe):** Independent application with integrated audio player.

2. USER INTERFACE

The interface is divided into three main sections:

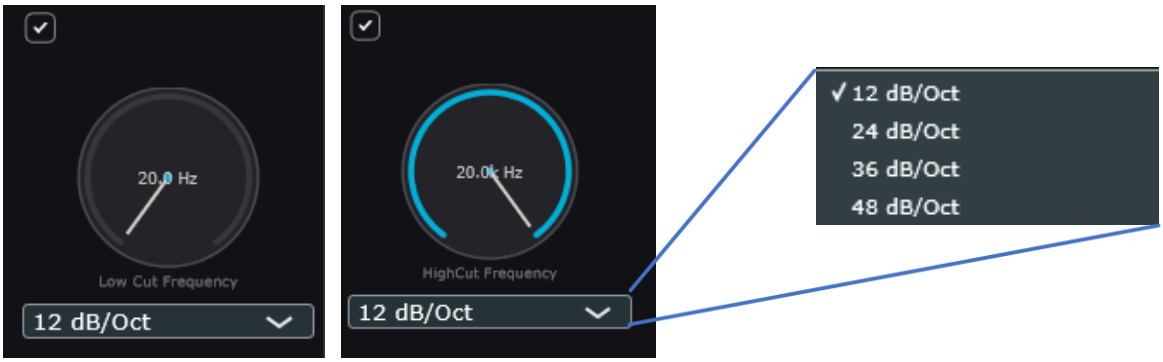
- **Controls Section (Lower):** Parameter settings by band.
- **Central Viewer:** Interactive spectrum and curve analyzer.
- **Utilities:** File player and default button.

3. EQUALIZER CONTROL

The plugin features 5 independent equalization bands:

Cut-off Bands (High-Pass and Low-Pass Filters)

Control	*Parameter*	*Description*
Low Cut Frequency	Frequency (Hz)	Bass attenuation point.
High Cut Frequency	Frequency (Hz)	Treble attenuation point.
Slope	Slope (dB/Oct)	Slope selection: 12, 24, 36 or 48 dB/Oct.

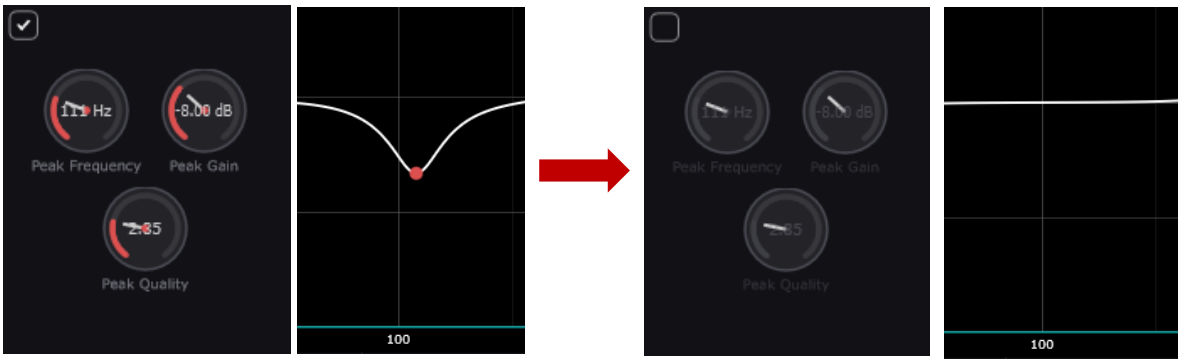


Parametric Bands (Peak Filters)

Control	*Parameter*	*Description*
Peak Frequency	Frequency (Hz)	Selection of the center frequency.
Peak Gain	Gain (dB)	Boost or attenuation (± 24 dB).
Peak Quality	Q Factor	Bell bandwidth.

4. ACTIVATION SYSTEM (ON/OFF) AND BYPASS

Each of the 5 bands has an independent activation button located at the top of its control column.

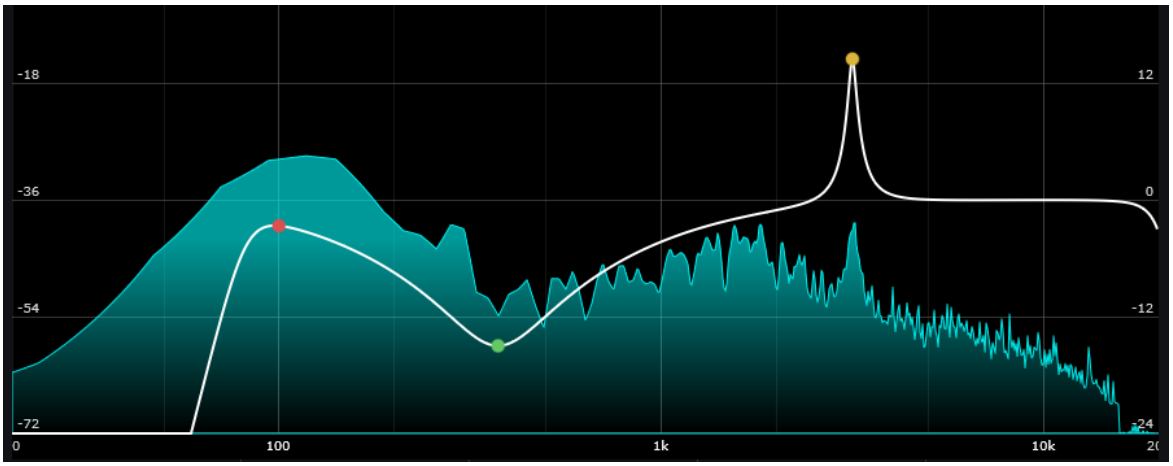


- **On/Off Function:** These switches allow you to instantly activate or deactivate (Bypass) the processing of each filter.

- **Parameter Memory:** When a filter is deactivated, the plugin retains the potentiometer settings (frequency, gain, and Q). This allows for quick A/B comparisons to evaluate the impact of each band on the sound without losing the adjustments made.

- **Visual Status:** The equalizer uses an intelligent attenuation system. When a filter is turned off, the corresponding potentiometers and labels reduce their opacity and color. This visual feedback allows you to identify at a glance which bands are processing audio and which are at rest.

5. CURVE VIEWER AND SPECTRUM ANALYZER



Dynamic Response Curve

The viewer functions as a real-time response monitor. The colored nodes on the screen serve as visual guides that automatically move according to the values the user adjusts in the aforementioned controls.

- **White curve:** represents the total sum of all active filters.

- **Nodes:** indicate the exact position (frequency and gain) of each Peak filter, allowing immediate visual validation of the applied processing.

Spectrum Analyzer

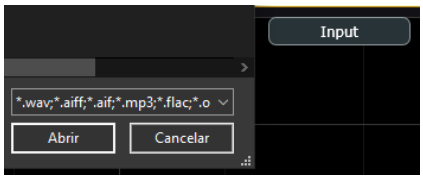
It incorporates a 2048-point FFT (Fast Fourier Transform) that maps the energy of the background audio signal. This allows you to visually identify resonances or energy gaps to help you decide which parameters to adjust in the lower controls.

6. GLOBAL CONTROLS

Default button

Resets all filters to their default position, in which no filtering is performed on the input

Audio playback controls

Standalone mode allows you to process external audio files without the need for a DAW:

- **Load Audio:** Opens a file explorer to load WAV, MP3, AIFF, etc. files.
- **Play Button:** Play/Stop controls for listening.
- **Gain Slider:** Adjusts the input level of the loaded track.